

DEVANG PATIL

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PROFESSIONAL SUMMARY

Passionate and systems-focused Computer Science student with hands-on experience building AI-powered applications, multi-agent workflows, and full-stack software architectures. Proficient in Python, REST APIs, and Large Language Models (LLMs), with practical experience implementing Prompt Engineering and cooperative agent structures. Skilled in databases (PostgreSQL, Supabase) and developer tools (Git/GitHub, Docker) to deliver scalable solutions.

TECHNICAL SKILLS

- **Languages:** Python (Intermediate), JavaScript, TypeScript, Rust, Go, C/C++, Kotlin
- **Artificial Intelligence:** LLMs (Gemini, Claude, OpenAI), Cooperative AI Agents, Prompt Engineering, Retrieval-Augmented Generation (RAG) Concepts
- **Backend & APIs:** REST APIs (FastAPI, Express), JSON Processing, SQL, PostgreSQL, Supabase, MySQL, MongoDB, SQLite
- **Frontend & Web:** React, Next.js, Tailwind CSS, Framer Motion, HTML5, Vanilla CSS
- **Tools & DevOps:** Git, GitHub, Docker, GitHub Actions, Render API, Vercel

EDUCATION

Vishwakarma Institute of Technology (VIT), Pune

Bachelor of Technology (B.Tech) in Computer Science

Pune, India

2025 — 2029

* **Core Coursework:** Object-Oriented Programming (OOP) in Python & Java, Data Structures & Algorithms, Systems & Operating Systems, Database Management Systems (DBMS), System Design.

PROJECTS

Singular — Cooperative AI Agent Platform

React, Vite, Supabase, Gemini API, Render API, REST APIs, JSON

* Designed and integrated **cooperative AI agent nodes** using the **Gemini API** to automate invoicing, bookkeeping, and receipt auditing for solopreneurs. * Engineered an **Invoice Sentinel** chatbot using custom **Prompt Engineering** to parse invoice contents, extract metadata in JSON format, and flag discrepancies. * Constructed a secure database backend on **Supabase** with relational tables, managed authentication, and hosted the application on **Render** with automatic CI/CD.

EcosystemAI — Biophysical Digital Twin & Socratic AI Coach

Next.js, Vercel, Gemini API, Aurora DSQL, REST APIs, JSON

* Created a **Socratic AI Coach** powered by the **Gemini API** that dynamically tailors prompts and explanations based on real-time quiz results and student interaction history. * Engineered a real-time predator-prey dynamics simulator modeled on Lotka-Volterra differential equations, allowing parallel simulation experiments. * Implemented a serverless relational database layer using **Aurora DSQL** to guarantee ACID transactions for concurrent classroom runs.

ChainCampus — Decentralized Identity & Academic Management System

React, Rust, Anchor, SQLite, Android SDK, NFC API, WebSockets, Git

* Programmed custom **Rust/Anchor smart contracts** to register on-chain student accounts, attendance logs, and verified scholarship transactions. * Built an **NFC Login Relay** enabling physical student card scanning via Android mobile devices to authenticate active desktop sessions securely through a WebSocket loop. * Developed a responsive React interface utilizing Context API for clean global state management and custom-rendered 3D interactive ID cards (Framer Motion).

AeroMesh 3D — Swarm Telemetry Control Engine & Real-Time 3D Cockpit

Next.js, React Three Fiber, Three.js, FastAPI, Python, WebSockets

* Engineered pathfinding algorithms utilizing **Artificial Potential Fields (APF)** and tangential vortex forces for drone deconfliction. * Broadcast telemetry at **20Hz** over **WebSockets** to three-dimensional control HUDs and interactive flight controllers. * Developed dynamic **3D viewport cockpits** using React Three Fiber, supporting interactive target markers and obstacle spawning.

EXPERIENCE & ACTIVITIES

Vishwaconclave — Research Volunteer

National Student Symposium

Pune, India

2025 — Present

* Helped research for prospective event speakers and compiled detailed dossiers. * Designed and formatted PowerPoint presentation decks used during live speaker sessions.